

MINISTRY OF EDUCATION AND TRAINING EASTERN INTERNATIONAL UNIVERSITY	SOCIALIST REPUBLIC OF VIETNAM Independence - Freedom - Happiness
	Binh Duong, March 8, 2026

PROJECT PROPOSAL

PART I. GENERAL INFORMATION

1.1 Research project title:

Bridging the Domain Gap for End-to-End Autonomous Driving in CARLA: From Structured Traffic to Unstructured Mixed Traffic.

1.2 Code (provided once the proposal is approved):.....

1.3 Project Status

☒ New ☐ Continuation of the research from the same research group

☐ Continuation of another's research

1.4 Field/Research Priority Areas (Research direction)

Priority Field 1: Computer Science; Research Direction: Artificial Intelligence

1.5 Type of Research

☐ Basic Research

☒ Applied Research

☐ Development Research

1.6 Research Field:

☐ Natural Sciences;

☐ Humanities..

☒ Engineering and Technology;

☐ Medical and Health Sciences.

☐ Social Sciences;

☐ Other

1.7 Proposed Duration: 9 months..

1.8 Total Budget

(Note: Ensure consistency between this section and Section 2.9 – Total Proposed Budget)

Total project budget: xxx VND. Includes:

No.	Funding Source	Amount (VND)	Percentage (%)
01	Funding provided by the University	0	
02	Self-funding	0	
03	Other funding sources (specify source: equity, sponsorship,	0	

	etc.)		
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(Attach documentation to verify funding sources, if applicable. Indicate the name of the sponsoring organization, if any)

1.9 Principal Investigator

1.9.1. Student (Principal Investigator)

Full Name: Phan Hoàng Kha

Date of Birth: 31/05/2005 Gender: Male ☒ / Female ☐

Student ID number: 2331200042

Faculty: School of Computing and Information Technology

Personal Tax Code:

Bank Account Number:.....Bank name:

Contact Address: 249/97 Le Hong Phong, Phu Loi Ward, Ho Chi Minh City

Phone Number: 0911928895 Email: kha.phan.cit23@eiu.edu.vn

1.9.2. Supervising Lecturer

Full Name: Phạm Đại Xuân

Date of Birth: 09/08/1971..... Gender: Male

Academic title:..... Degree: Ph.D.

Faculty/Department: School of Computing and Information Technology

Position: Instructor

Personal Tax Code:

Bank Account Number:..... Bank:

Contact Address: 91/18B Le Van Duyet, Gia Dinh Ward, HoChiMinh City

Phone Number: 0909156924Email: xuan.phamdai@eiu.edu.vn

1.10 Summary of Relevant Experience Related to the Research Group's Project

1.11 Organization

Name of the Organization: Eastern International University

Address: Nam Ky Khoi Nghia street, Hoa Phu ward, Thu Dau Mot city, Binh Duong province, Vietnam

Phone Number: (+84) 0274 222 0372

Tax Code: 3701807809

Bank Name and Address: Joint Stock Commercial Bank for Investment and Development of Vietnam - Binh Duong Branch (BIDV Binh Duong), 549 Binh Duong Avenue, Hiep Thanh Ward, Thu Dau Mot City, Binh Duong Province, Vietnam

Account Name: Trường Đại học Quốc tế Miền Đông

Account number: 65010000556422

Representative: Dr. Ngo Minh Duc

Position: President

1.12 Research Personnel

(Include individuals who contribute scientifically and perform key tasks within the lead and collaborating organizations involved in the project.)

No.	Full name (with title, degree)	Affiliation	Assignment of tasks (state the serial number of the assigned task)	Working time on the project (Number of working days)
I	Principal investigator:			
1	Phan Hoàng Kha	School of Computing and Information Technology	1,2,3,4	100
II	Member			
1	Lê Thiện Thành	School of Computing and Information Technology	1,2,3,4	100
2	Ngô Thị Kim Hoa	School of Computing and Information Technology	1,2,3,4	100
3	Hoàng Thị Thảo Vân	School of Computing and Information Technology	1,2,3,4	100
4	Ph.D. Phạm Đại Xuân	School of Computing and Information Technology	5	

1.13 Main collaborating organizations in implementing the project (if any)

(Collaboration confirmation form A4-PHNC.SV)

1.13.1 Organization 1:

Name of managing agency:

Phone number: Fax:

Address:

Full name of the head of the organization:

Bank account number:Bank name:

Capacity to contribute to the project (maximum 500 characters):

1.13.2 Organization 2:

Name of managing agency

Phone number: Fax:

Address:

Full name of the head of the organization:

Bank account number:Bank name:

Capacity to contribute to the project (maximum 500 characters)

PART II: RESEARCH DESCRIPTION

2.1 Overview of the research situation, both domestically and internationally

(Based on the assessment of the research situation domestically and internationally, analyze the ongoing research processes, the latest results related to the topic, evaluate the differences between the current state of science and technology domestically and globally, the issues that have been resolved, and clearly state the existing issues.)

Internationally, autonomous vehicles have made significant progress in recent years, particularly with the development of machine learning models and high-fidelity simulators for autonomous vehicle research. Many existing autonomous driving systems are developed and trained on structured traffic scenarios (featuring traffic lights, clear lane markings, and predominantly four-wheeled vehicles).

In Vietnam, notable research such as that by Phenikaa Group (2021) has demonstrated the ability to master Level 4 autonomous vehicle technology within closed campus environments. However, the traffic environment in many developing countries including Vietnam is characterized by unstructured, mixed traffic with a diverse range of road users and unpredictable driving behaviors.

2.2 Scientific ideas, relevance, and innovation

(Only specify the limitations in the current level of science and technology domestically and globally, from which new solutions are proposed – discuss the objectives set for the topic and its relevance, the benefits of research results for the industry, the organization leading the project, and society.)

Idea: This research will focus on analyzing the domain gap between structured environments and unstructured mixed environments for autonomous driving systems and autonomous driving algorithms. The purpose of this study is to evaluate the performance

of autonomous driving models trained using structured traffic scenarios when applied to unstructured, mixed traffic environments [1].

Innovation: Investigating the performance of autonomous vehicles in unstructured, mixed traffic scenarios, where multiple traffic participants interact simultaneously. The research also aims to explore domain adaptation methods to help the driving system adapt to more complex traffic environments.

2.3 Preliminary research results

(Prior to submitting this outline, the research group may have conducted preliminary studies; if so, present the results and the techniques used.)

Successfully set up the CARLA simulator and tested the sensors (including building the simulator from source).

Established a data collection pipeline: successfully extracted sensor data from CARLA, including RGB Camera and LiDAR feeds.

Conducted trials spawning diverse actors via the CARLA Traffic Manager (e.g., pedestrians, bicycles, and motorcycles) to validate the capability of generating a mixed-traffic environment.

2.4 References

(Name of the work, author, place and year of publication, only list the references cited in this proposal)

[1] A. Farhadi, M. Mirzarezaee, A. Sharifi, and M. Teshnehlab,

“Domain adaptation in reinforcement learning: A comprehensive and systematic study,”

Frontiers of Information Technology & Electronic Engineering,
vol. 25, no. 11, pp. 1446–1465, 2024.

[2] A. Dosovitskiy, G. Ros, F. Codevilla, A. López, and V. Koltun,

“CARLA: An Open Urban Driving Simulator,”

in Proc. 1st Conf. Robot Learning (CoRL), 2017.

2.5 Research objectives and content

2.5.1 Objectives

(Clearly state the scientific and practical objectives that the project aims to address, and the level to which they will be solved – Align with the needs of the intended sector or customer – if applicable.)

Research and develop a highly adaptable End-to-End (E2E) autonomous driving model to bridge the domain gap when transitioning from structured traffic environments to unstructured, mixed-traffic environments within the CARLA simulator.

Establish a Baseline Model: Successfully implement and train an End-to-End autonomous driving system that operates reliably in structured traffic environments (characterized by clear lane markings, distinct traffic signals, and predictable behaviors).

Analyze and Quantify the Domain Gap: Evaluate the extent of performance degradation and identify safety risks when deploying the baseline model in unstructured traffic environments (characterized by a lack of lane markings, alongside complex and unpredictable trajectories of surrounding vehicles).

Implement Domain Adaptation: Apply appropriate domain adaptation techniques to enhance the stability and robustness of the autonomous vehicle when navigating mixed-traffic scenarios.

Evaluate Safety and Performance: Comprehensively assess the safety, situational handling capabilities, and overall reliability of the optimized model using the standardized evaluation metrics of the CARLA Leaderboard.

2.5.2 Research Content and Methods

(List and describe in detail the content and methods of the research, clearly explaining the experiments and elaborating on activities to justify the budget appendix.)

a) Content 1: Successfully Deploying an End-to-End (E2E) Autonomous Vehicle in a Structured Environment

- Task 1: Review current End-to-End (E2E) models, focusing on Imitation Learning and Reinforcement Learning. Select an appropriate baseline model.
- Task 2: Implement the E2E algorithm.
 - + **Methodology:** Collect data within structured CARLA maps using the AutoPilot agent. Train the E2E model (Reinforcement Learning)
 - + Evaluate the route completion rate.

b) Content 2: Establishing an Unstructured, Mixed-Traffic Environment in CARLA

- Task 1: Customize the CARLA environment to simulate complex traffic scenarios typical of developing countries.

c) Content 3: Domain Gap Analysis and Domain Adaptation Application

- Task 1: Deploy the baseline model trained in Content 1 into the unstructured environment established in Content 2. Analyze the model's performance degradation to quantify the domain gap.

- Task 2: Apply appropriate Domain Adaptation techniques to bridge the identified gap.

d) Content 4: Safety Assessment and Performance Analysis

- Task 1: Establish safety evaluation criteria. Utilize standardized metrics from the CARLA Leaderboard (e.g., collision rate, route completion, traffic light infractions) to analyze the system's safety.
- Task 2: Conduct comprehensive model evaluation. Test the adapted model across the specific unstructured scenarios outlined in Content 2.

2.5.3 Summary of Tasks, Implementation Timeline, and Assignments

No.	Main Tasks to be Completed	Expected Scientific Outcomes and Evaluation Criteria	Timeline (from 4/2026 to the end of 12/2026)	Responsible Individual(s) or Organization(s)	Estimated Budget (VND)
(1)	(2)	(3)	(4)	(5)	(6)
1	Content 1: Successfully Deploying an End-to-End (E2E) Autonomous Vehicle in a Structured Environment	Successfully run E2E AV in structured environment	From 4/2026 to 6/2026	Phan Hoàng Kha, Lê Thiện Thành, Hoàng Thị Thảo Vân, Ngô Thị Kim Hoa	
1.1	Task 1: Review current End-to-End (E2E) models, focusing on Imitation Learning and Reinforcement Learning. Select an appropriate baseline model.	Select and apply appropriate algorithm	From 1/4/2026 to 1/5/2026	Phan Hoàng Kha, Lê Thiện Thành, Hoàng Thị Thảo Vân, Ngô Thị Kim Hoa	
1.2	Task 2: Implement the E2E algorithm.	Refine and optimise selected	From 1/5/2026 to	Phan Hoàng Kha, Lê Thiện Thành, Hoàng	

		algorithm	31/5/2026	Thị Thảo Vân, Ngô Thị Kim Hoa	
2	Content 2: Establishing an Unstructured, Mixed-Traffic Environment in CARLA	Obtain unstructure d environmen t in Carla	From 6/2026 to 9/2026	Phan Hoàng Kha, Lê Thiện Thành, Hoàng Thị Thảo Vân, Ngô Thị Kim Hoa	
2.1	Task 1: Customize the CARLA environment to simulate complex traffic scenarios typical of developing countries.	Obtain customised unstructure d environmen t to simulate complex traffic scenarios	From 1/6/2026 to 31/8/2026	Phan Hoàng Kha, Lê Thiện Thành, Hoàng Thị Thảo Vân, Ngô Thị Kim Hoa	
3	Content 3: Domain Gap Analysis and Domain Adaptation Application	Successfully apply domain adaptation method	From 9/2026 to 12/2026	Phan Hoàng Kha, Lê Thiện Thành, Hoàng Thị Thảo Vân, Ngô Thị Kim Hoa	
3.1	Task 1: Deploy the baseline model trained in Content 1 into the unstructured environment established in Content 2. Analyze the model's performance degradation to quantify the domain gap.	Successfully analyse model performanc e	From 1/9/2026 to 31/10/202 6	Phan Hoàng Kha, Lê Thiện Thành, Hoàng Thị Thảo Vân, Ngô Thị Kim Hoa	
3.2	Task 2: Apply appropriate Domain Adaptation	Apply domain adaptation	From 1/11/2026 to 30/11/202	Phan Hoàng Kha, Lê Thiện Thành, Hoàng Thị Thảo Vân,	

	techniques to bridge the identified gap.		6	Ngô Thị Kim Hoa	
4	Content 4: Safety Assessment and Performance Analysis	Successfully evaluate safety and credibility based on CARLA leaderboard	From 11/2026 to 12/2026	Phan Hoàng Kha, Lê Thiện Thành, Hoàng Thị Thảo Vân, Ngô Thị Kim Hoa	
4.1	Task 1: Establish safety evaluation criteria. Utilize standardized metrics from the CARLA Leaderboard (e.g., collision rate, route completion, traffic light infractions) to analyze the system's safety.	Successfully construct safety assessment criterias	From 1/12/2026 to 15/12/2026	Phan Hoàng Kha, Lê Thiện Thành, Hoàng Thị Thảo Vân, Ngô Thị Kim Hoa	
4.2	Task 2: Conduct comprehensive model evaluation. Test the adapted model across the specific unstructured scenarios outlined in Content 2.	Obtain statistics from evaluation process	From 15/12/2026 to 30/12/2026	Phan Hoàng Kha, Lê Thiện Thành, Hoàng Thị Thảo Vân, Ngô Thị Kim Hoa	
5	Monitor, assist, ensure data integrity	Work completion on schedule	From 4/2026 to 31/12/2026	Ph.D. Phạm Đại Xuân	

2.5.4 Feasibility

a) Regarding raw materials and energy

Content	No.	List of Required Materials and Energy	Funding Source* (From EIU; etc)
Content 1	01	Laptop or personal computer	Self-funding
	02	Electricity for powering devices	EIU
	03	Internet connection for data gathering & testing	EIU
Content 2	05	Python IDEs (e.g., VS Code, PyCharm) Online platform (Kaggle, Google Colab)	Free
Content 3	06	Server hosting for online prototype	EIU
Content 4	07	Storage for datasets and model files (local or cloud)	EIU

**For funding from Eastern International University, provide justification in Appendix Section 2; written documentation is required for funding from other sources.*

b) Regarding equipment

Content	No.	List of required equipment	Capability to mobilize	
			Currently available at EIU	Not available at EIU* (Outsourcing; utilizing from partners; new purchases from which funding sources, etc.)
Content	1	Laptop or PC for ML	AI Lab	

1, 2		development		
	2	Web server for hosting prototype	AI Lab	
Content 3,4	1	High-performance computer	AI Lab	

**For equipment not available at EIU: if funding is sourced from EIU, provide an explanation in Appendix Section 4; provide written evidence for funding from other sources.*

c) Regarding Research Team

(Clarify the capability and experience of the research team and partners to affirm the feasibility of each proposed research content)

Content	Capability and experience of the research team	Capability and experience of the partners

d) Plan for Collaboration with Research Organizations and Production Facilities (Domestic and International)

(Please clearly state the collaboration plan: names of the main collaborating organizations participating in the project and the scope of their involvement, including production facilities or end users of the research results; and their potential contributions in terms of personnel, financial support, and infrastructure—if applicable).

2.6 Research results

2.6.1 Description of Products/Research Outcomes (Required)

- Published Products *(including articles, books, textbooks, guides, etc.)*

No .	Product Name	Quantitative Evaluation Criteria	Note

- Applied Products *(including patents, inventions, designs, scientific data, models, equipment, products, software, etc.)*

No.	Main Quality Indicators of the Product	Unit of Measurement	Quality Level - Quantitative Evaluation Criteria	Estimated Quantity/Scale of Product Output
1	End-to-End autonomous driving model/algorithm (Integrated with Domain Adaptation method)	Model	Stable operation within CARLA. The route completion rate in unstructured environments increases by at least 10-15% compared to the baseline model (the original model without adaptation).	1
2	Mixed-traffic simulation dataset (Contains sensor data: RGB Camera, LiDAR, etc.)	GB	Includes diverse unstructured traffic scenarios (e.g., motorcycles, jaywalking pedestrians). Data is fully labeled and time-synchronized across all sensors.	~50GB
3	Test Scenarios	Scenarios	Simulates complex traffic scenarios.	Complex scenarios combined by conditions (motorcycles, jaywalking pedestrians, ...).

- Other Products (including the final project report, summary report, if applicable, ...)

No.	Product name	Quantitative Evaluation Criteria	Note

2.6.2 Academic Contributions (Application of research project outcomes in course results/thesis/graduation dissertations)

No .	Education Level	Quantity <i>(Number of students)</i>	Field of Study	Note
01	Undergraduate			
02	Master's			
03	Doctoral			

2.7 Research Contributions

2.7.1 New contributions to knowledge; the extent to which the research problem is addressed

- Development of a Specialized Dataset for Unstructured Traffic
- End-to-End (E2E) Autonomous Driving Architecture
- Theoretical Framework for Domain Adaptation

2.7.2 Practical contributions and the potential for real-world application

- Standardized Safety Benchmarking via CARLA Leaderboard
- Operational Safety and Reliability Assessment

2.7.3 Development of the research group

(Comparison of the group's expertise with domestic and international groups in the same field)

2.7.4 Potential for transferring research outcomes

(Applicable for deployment research types)

2.8. Plan for managing assets resulting from the implementation of the research project *(Methods of management and beneficiaries)*

2.9. Summary of Proposed Funding

(Detailed explanation provided in the attached appendix)./.

Confirmation From the Leader of School/ Department	Principal Investigator Phan Hoàng Kha Supervising Lecturer
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	Phạm Đại Xuân
President	Committee Chair

PART III: APPENDIX
EXPLANATION OF EXPENSE ITEMS

Unit: VND

No.	ITEM DESCRIPTION	Unit	Quantity	Unit Price	Amount	Funding Source		Notes
						EIU	Other (.....)	
I	Materials, Supplies, and Consumables							
1								
2								
II	Field Investigation and Data Collection (e.g., number of forms, expenditure/form)	Form						
III	Participation in Scientific Conferences							
1	International conferences held abroad							
-	Southeast Asian countries	Time(s)						
-	Other countries	Time(s)						
2	International conferences held domestically	Time(s)						
3	Domestic scientific conferences	Time(s)						
IV	Publication in Scientific Journals Listed by NAFOSTED							
1	International publications	paper						
2	Domestic publications	paper						
V	Inventions, Patents, and Scientific Products Registered for Intellectual Property Rights							
1	Domestic intellectual property registration							
-	Patent	certificate						
-	Other IP rights (utility solutions, industrial designs, trademarks, etc.)	certificate						
2	Overseas intellectual property registration	certificate						
VI	Services Supporting Research Activities							
1								
Total Budget					-			

